

Introduction to BJTU-RAO Bogie Datasets

1. Datasets overview

BJTU-RAO bogie datasets are provided by the **State Key Lab of Advanced Rail Autonomous Operation at Beijing Jiaotong University (BJTU)**, Beijing, P.R. China. The datasets contain multi-sensor data of transmission systems of subway train bogies corresponding to 51 health states that were acquired by fault simulation experiments. The BJTU-RAO bogie datasets are **the first publicly available fault datasets of train transmission systems worldwide**, aiming to encourage scholars and engineers to study and validate their fault detection or diagnosis algorithms for key components of train transmission systems, including traction motors, driving gearboxes and axle boxes.

These datasets are publicly available, and anyone can feel free to use them when the data source is indicated. Publications making use of the BJTU-RAO bogie datasets are requested to cite the following paper.

A. Ding, Y. Qin, B. Wang, L. Guo, L. Jia, and X. Cheng, Evolvable graph neural network for system-level incremental fault diagnosis of train transmission systems, *Mechanical Systems and Signal Processing*, 210 (2024) 111175.

2. Background

The safety and reliability of trains have always been the top priority of the railway industry, and now more than ever it is essential to develop new solutions to meet huge challenges from higher operation speed and operation intensity. The transmission systems of train bogies, including traction motors, driving gearboxes and axle boxes, can transfer kinetic energy from motors to wheelsets, as shown in Fig. 1. Their health condition will directly affect the operation safety of trains. Therefore, it is vital to realize real-time fault diagnosis of train transmission systems for ensuring safe and reliable operations.

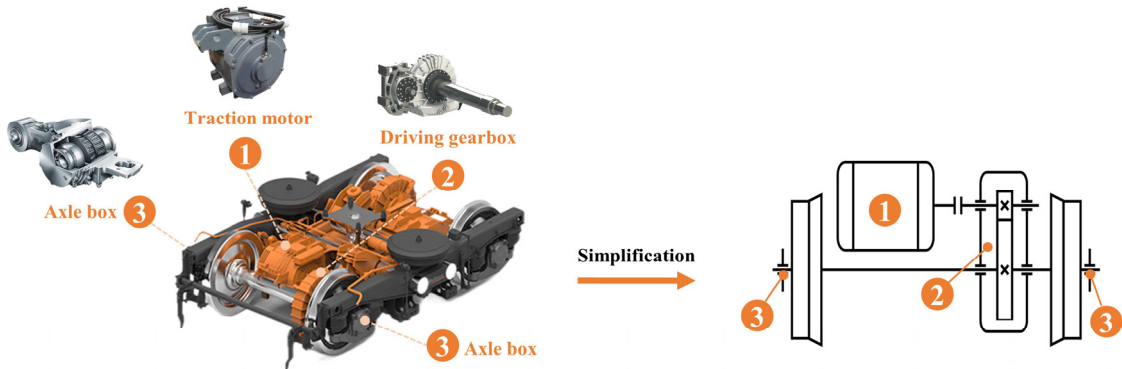


Fig.1 Schematic diagram of transmission systems in the train bogies.

3. Dataset characteristics

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 The datasets are acquired from an experimental platform designed based on **an isotropic downsizing bogie of real subway trains**.
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 The datasets cover **24-channel sensor data** of vibration, current, rotational speed and sound intensity.
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51 health states of transmission systems of train bogies are contained in the datasets, including typical single faults, component-level compound faults and system-level compound faults.
- 
 The sensor deployment is highly reductive to that on a real train, and different working condition settings are considered to simulate the various operating conditions of trains.

4. Brief introduction to experiments

4.1 Experimental platform

The datasets are collected from the experimental platform for fault simulation of subway train bogies, as shown in Fig. 2. This platform is designed and adjusted according to the real bogies of subway trains, and the scale of the experimental platform to the real bogie is 1:2. The single power transmission chain includes a motor, a gearbox and two axle boxes. The transmission chain is driven by a three-phase asynchronous motor, and the motor speed is controlled via a frequency inverter. The load is applied by means of an electro-hydraulic load device. The motor bearing type is SKF 6205-2RSH, and the gears of the reduction gearbox are helical gears. The number of the driving gear teeth is 16, while the number of the driven gear teeth is 107. The axial bearing type of driving gear is HRB 32305. The axle box bearing type is HRB 352213. The fault characteristic frequency of these bearings can refer to Table 1.

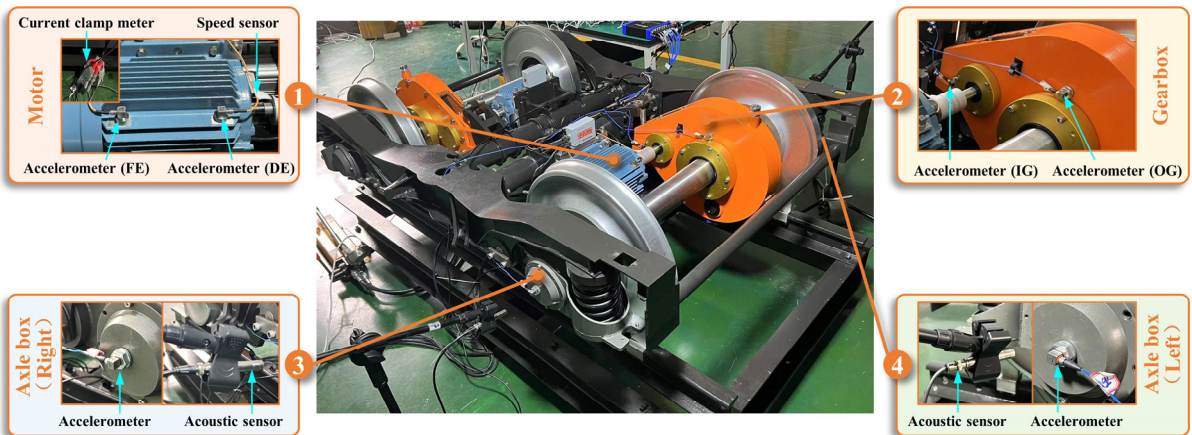


Fig.2 Experimental platform of subway train bogies.

Table 1 Fault characteristic frequency of bearings.

Type	Fault Characteristic Frequency (Rotation Frequency f)			
	Inner Ring	Outer Ring	Rolling Element	Cage
SKF 6205-2RSH	$5.415*f$	$3.585*f$	$2.357*f$	$0.398*f$
HRB 32305	$7.889*f$	$5.111*f$	$2.203*f$	$0.393*f$
HRB 352213	$10.929*f$	$8.071*f$	$3.164*f$	$0.425*f$

4.2 Working conditions

Samples under **9 different working conditions** are given for each type of health state in the BJTU-RAO bogie datasets that can represent the different operating conditions of trains, as listed in Table 2. Different motor speed is to simulate different train speed, while different lateral load is to simulate trains going straight or round corners. The positive value of load denotes that the loading direction points to the motor side. Otherwise, it denotes that the loading direction points to the gearbox side.

Table 2 Working conditions in BJTU-RAO bogie datasets.

Working condition	Motor speed/lateral load	Working condition	Motor speed/lateral load
WC1	20Hz/0kN	WC6	60Hz/+10kN
WC2	40Hz/0kN	WC7	20Hz/-10kN
WC3	60Hz/0kN	WC8	40Hz/-10kN
WC4	20Hz/+10kN	WC9	60Hz/-10kN
WC5	40Hz/+10kN		

4.3 Sampling setting

The BJTU-RAO bogie datasets contain **four kinds of signals**, i.e. three-axis vibration acceleration, three-phase current, rotational speed, and sound intensity, with **a total of 24 channels**. The sampling frequency of all sensors is **64kHz**. The deployment of sensors is highly reproducible to the real trains (see Fig. 2), and the descriptions of sensor channels are given in Table 3.

Table 3 Summary of sensor channel information.

Channel	Component	Deployment location	Signal type	Unit
CH1	Motor	Motor (drive end)	Tri-axial acceleration	g
CH2				g
CH3				g
CH4		Motor (fan end)	Tri-axial acceleration	g
CH5				g

Channel	Component	Deployment location	Signal type	Unit
CH6		Motor (cable)	Three-phase current	g
CH7				A
CH8				A
CH9				A
CH10		Motor (shaft)	Rotational speed	V
CH11	Gearbox	Gearbox (input axle)	Tri-axial acceleration	g
CH12				g
CH13				g
CH14		Gearbox (output axle)	Tri-axial acceleration	g
CH15				g
CH16				g
CH17	Axle box (left)	Axle box (end cover)	Tri-axial acceleration	g
CH18				g
CH19				g
CH20		Axle box (adjacency)	Sound intensity	Pa
CH21	Axle box (right)	Axle box (end cover)	Tri-axial acceleration	g
CH22				g
CH23				g
CH24		Axle box (adjacency)	Sound intensity	Pa

4.4 Fault types and sample labels

The basic health state codes of each component (i.e., motor, gearbox, and axle box) in the BJTU-RAO bogie datasets are shown in Table 4, and Fig. 3 shows the photographs of fault simulations for each component. Label examples are listed in Table 5, which explains the labels of typical single faults, component-level compound faults, and system-level compound faults based on basic health state codes. All of the health state labels in BJTU-RAO bogie datasets are summarized in Table 6.

Table 4 Health state codes of BJTU-RAO bogie datasets.

Component	Basic health state	Description
Motor	M0	Normal condition
	M1	Short circuit
	M2	Broken rotor bar

Component	Basic health state	Description
	M3	Bearing fault
	M4	Bowed axis
Gearbox	G0	Normal condition
	G1	Gear cracked tooth
	G2	Gear worn tooth
	G3	Gear missing tooth
	G4	Gear chipped tooth
	G5	Bearing inner race fault
	G6	Bearing outer race fault
	G7	Bearing rolling element fault
	G8	Bearing cage fault
Axle box (left)	LA0	Normal condition
	LA1	Bearing inner race fault
	LA2	Bearing outer race fault
	LA3	Bearing rolling element fault
	LA4	Bearing cage fault
Axle box (right)	RA0	Normal condition
	RA1	Bearing inner race fault

Table 5 Sample labeling examples.

Example		Fault description	Label
Single faults	Example 1	Short circuit (Motor)	M1_G0_LA0_RA0
	Example 2	Gear cracked tooth (Gearbox)	M0_G1_LA0_RA0
Component-level compound faults	Example 1	Gear chipped tooth (Gearbox) + Bearing inner race fault (Gearbox)	M0_G4+G5_LA0_RA0
	Example 2	Bearing inner race fault (Left axle box) + Bearing outer race fault (Left axle box) + Bearing rolling element fault (Left axle box) + Bearing cage fault (Left axle box)	M0_G0_LA1+LA2+LA3+LA4_RA0
System-level compound faults	Example 1	Short circuit (Motor) + Bearing inner race fault (Left axle box)	M1_G0_LA1_RA0
	Example 2	Gear missing tooth (Gearbox) + Bearing inner race fault (Left axle box)	M0_G3_LA1_RA0

Table 6 All labels in BJTU-RAO bogie datasets.

Group	Number	Label
Normal & single faults	1	M0_G0_LA0_RA0
	2	M1_G0_LA0_RA0
	3	M2_G0_LA0_RA0
	4	M3_G0_LA0_RA0
	5	M4_G0_LA0_RA0
	6	M0_G1_LA0_RA0
	7	M0_G2_LA0_RA0
	8	M0_G3_LA0_RA0
	9	M0_G4_LA0_RA0
	10	M0_G5_LA0_RA0
	11	M0_G6_LA0_RA0
	12	M0_G7_LA0_RA0
	13	M0_G8_LA0_RA0
	14	M0_G0_LA1_RA0
	15	M0_G0_LA2_RA0
	16	M0_G0_LA3_RA0
	17	M0_G0_LA4_RA0
Component-level compound faults	18	M0_G4+G5_LA0_RA0
	19	M0_G2+G5_LA0_RA0
	20	M0_G3+G5_LA0_RA0
	21	M0_G1+G5_LA0_RA0
	22	M0_G0_LA1+LA2+LA4_RA0
	23	M0_G0_LA1+LA2_RA0
	24	M0_G0_LA1+LA2+LA3+LA4_RA0
	25	M0_G0_LA2+LA4_RA0
	26	M0_G0_LA1+LA2+LA3_RA0
	27	M0_G0_LA2+LA3_RA0
System-level compound faults	28	M2_G0_LA1_RA1
	29	M2_G5_LA0_RA0
	30	M2_G3_LA0_RA0
	31	M1_G5_LA0_RA0
	32	M1_G3_LA0_RA0
	33	M4_G5_LA0_RA0

Group	Number	Label
	34	M4_G3_LA0_RA0
	35	M3_G3_LA0_RA0
	36	M3_G5_LA0_RA0
	37	M1_G0_LA1_RA0
	38	M2_G0_LA1_RA0
	39	M3_G0_LA1_RA0
	40	M1_G0_LA1_RA1
	41	M0_G0_LA1_RA1
	42	M0_G3_LA1_RA0
	43	M1_G5_LA1_RA1
	44	M4_G3_LA1_RA1
	45	M4_G3_LA1_RA0
	46	M4_G5_LA1_RA0
	47	M2_G3_LA1_RA0
	48	M2_G5_LA1_RA0
	49	M1_G3_LA1_RA0
	50	M1_G5_LA1_RA0
	51	M4_G0_LA1_RA0



Fig. 3. Photographs of fault simulations. (a) Motor. (b) Gearbox. (c) Axle box.

4.5 Folder structures

The structure of the dataset folder is shown in Fig. 4. There are **51 sub-folders** in the main folder named *BJTU_RAO_Bogie_Datasets*, which corresponds to the 51 health states. These sub-folders are named using the health state labels of train transmission systems. **Each sub-folder contains 9 samples**, corresponding to 9 working conditions. **Each sample is composed of 4 CSV files**, corresponding to the sensor data collected from the traction motor, the gearbox and two axle boxes. The CSV file is named in the format of *data_monitoring_object_health_state_label_motor_speed_lateral_load.csv*. It should be noted that **each sample contains 10-seconds data**. Fig. 5 shows an example of data in a CSV file, in which the columns indicate each sensor channel and the rows denote the sampling points. The descriptions of each channel have been given in Table 3.

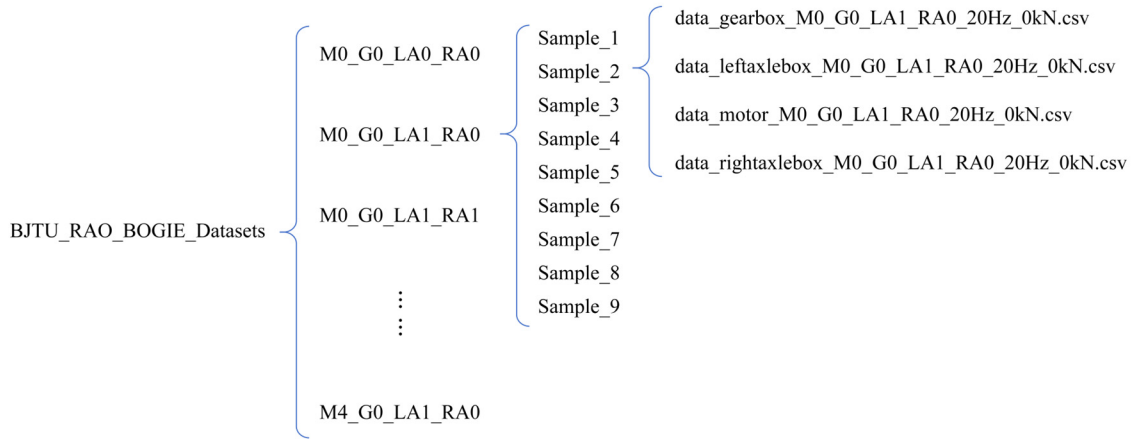


Fig.4 Folder structures of provided datasets.

	A	B	C	D	E	F	G
1	CH11	CH12	CH13	CH14	CH15	CH16	
2	-0.26545	0.163011	-0.30128	-0.45429	-0.23777	0.056932	
3	-0.21143	0.296719	-0.28026	-0.35464	-0.15132	0.11108	
4	-0.11571	0.394545	-0.21979	-0.30354	-0.12244	0.070156	
5	0.011392	0.454205	-0.13411	-0.19774	-0.07158	0.092628	
6	0.15831	0.473466	-0.08311	-0.10538	0.011648	0.116918	
7	0.303793	0.452841	-0.05014	0.00223	0.098963	0.178736	
8	0.42875	0.404659	-0.04653	0.098565	0.105284	0.205426	
9	0.500568	0.322713	-0.06925	0.119276	0.038622	0.155895	
10	0.528892	0.241875	-0.09822	0.18375	0.034773	0.217741	
11	0.497784	0.175938	-0.12148	0.237784	0.01821	0.268196	
12	0.449403	0.103324	-0.15575	0.194162	-0.05763	0.240369	
13	0.356605	0.035369	-0.18477	0.14	-0.11568	0.201548	
14	0.207045	-0.01516	-0.15797	0.114588	-0.15378	0.187202	
15	0.058935	-0.05774	-0.15852	0.083693	-0.11362	0.198949	
16	-0.07952	-0.09754	-0.15815	0.027798	-0.06923	0.159063	
17	-0.19541	-0.1406	-0.1556	-0.01017	-0.06143	0.111705	
18	-0.29023	-0.18977	-0.17892	0.042259	0.068665	0.122955	
19	-0.33246	-0.24898	-0.16141	0.115753	0.237699	0.136051	
20	-0.32388	-0.31683	-0.16271	0.191818	0.31544	0.098438	

Fig.5 An example of a CSV file containing sensor data of the motor.

4.6 Signal examples

Figs. 6(a)-(x) give examples of the time domain waveforms and envelope spectra for each channel of one sample, in which only 2-second data are intercepted to plot figures.

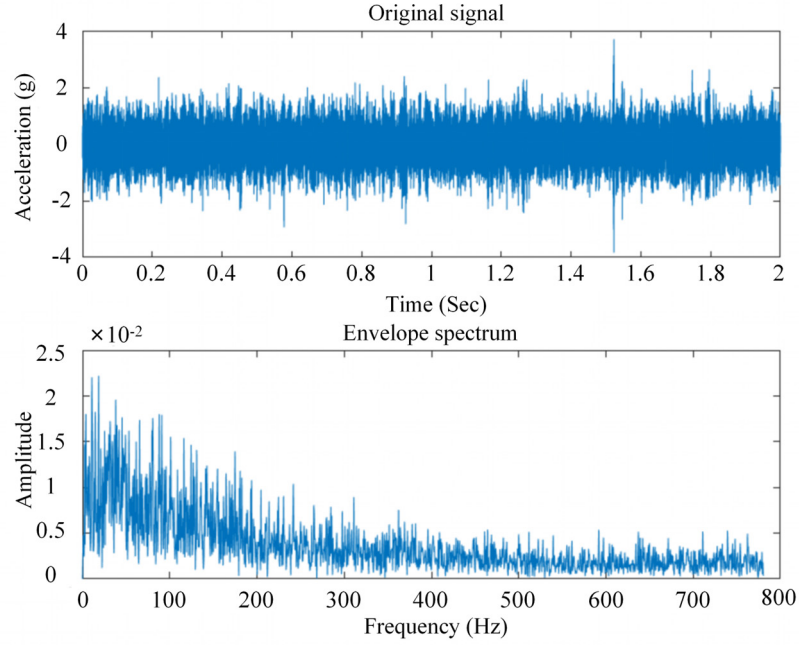


Fig. 6(a) Time domain waveform and envelope spectrum of Channel 1.

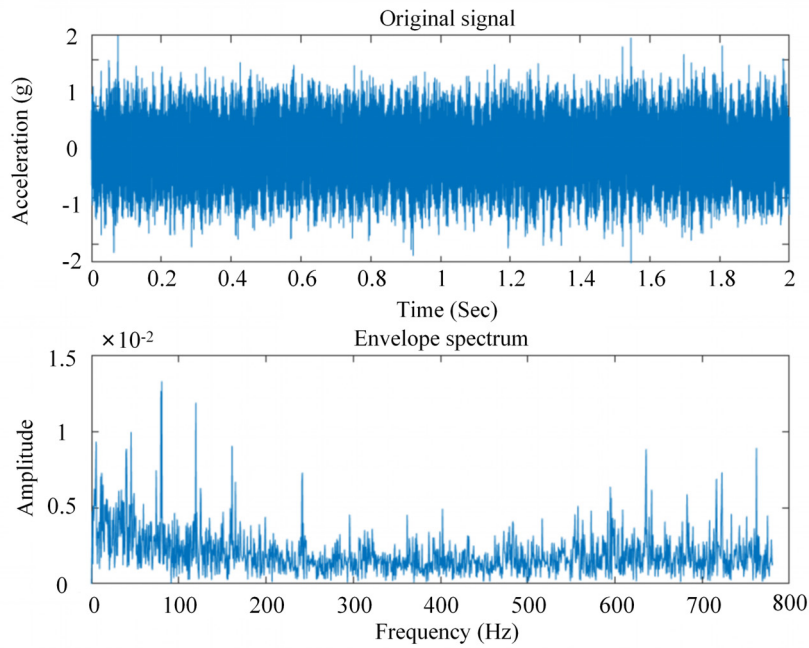


Fig. 6(b) Time domain waveform and envelope spectrum of Channel 2.

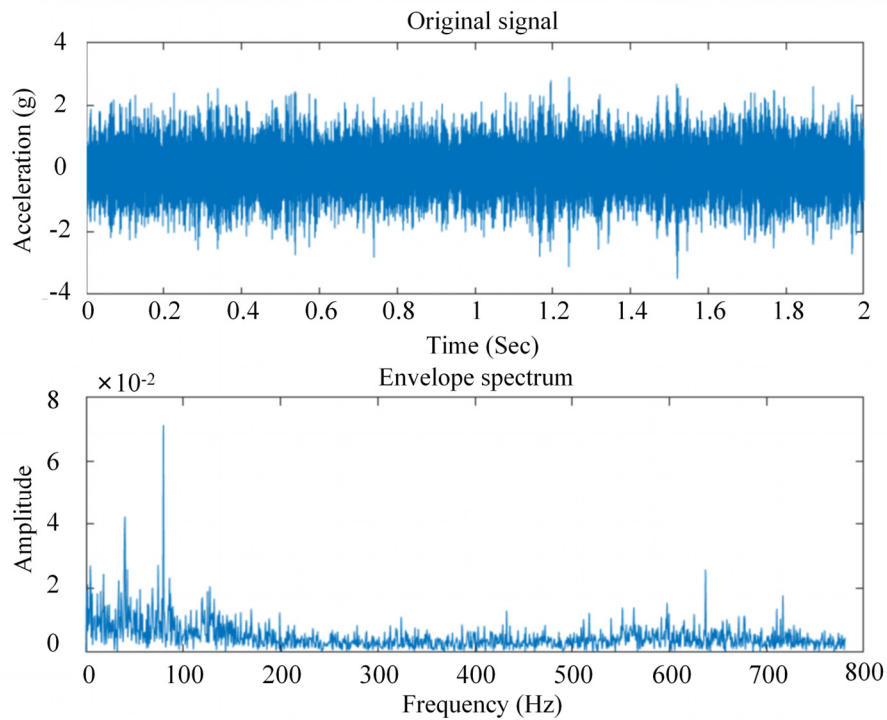


Fig. 6(c) Time domain waveform and envelope spectrum of Channel 3.

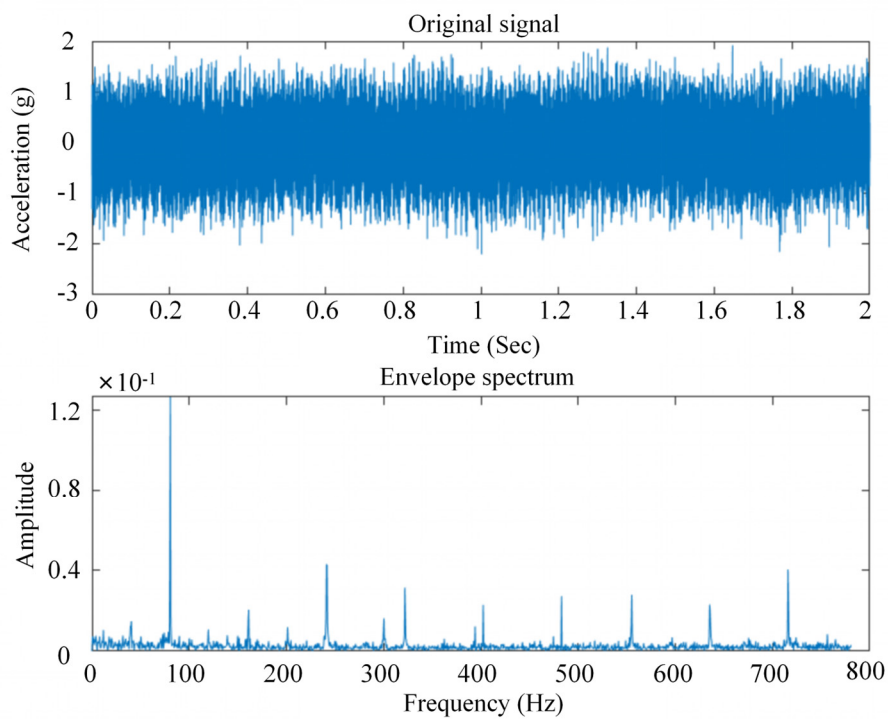


Fig. 6(d) Time domain waveform and envelope spectrum of Channel 4.

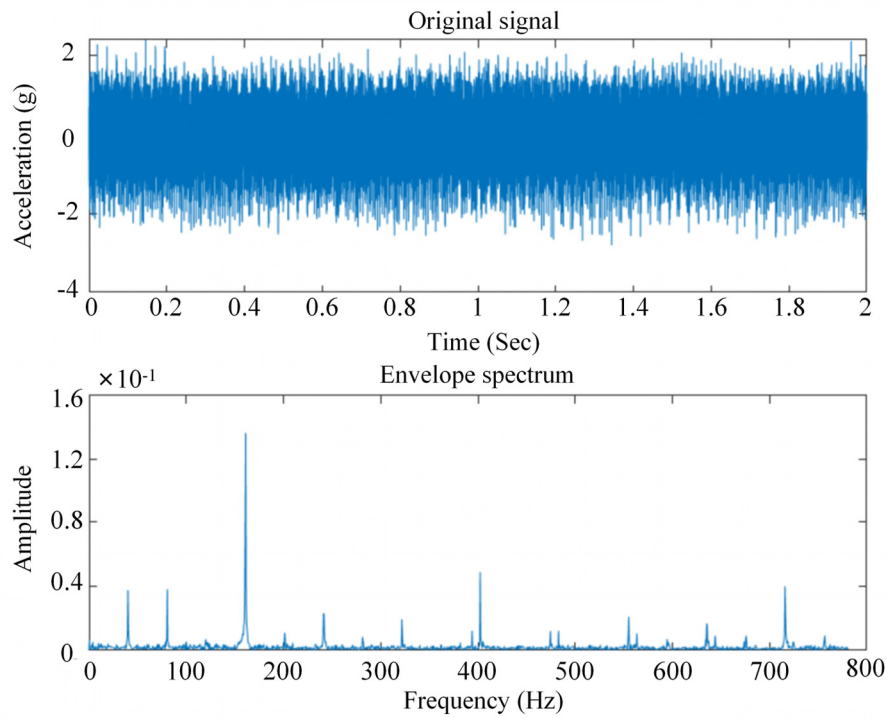


Fig. 6(e) Time domain waveform and envelope spectrum of Channel 5.

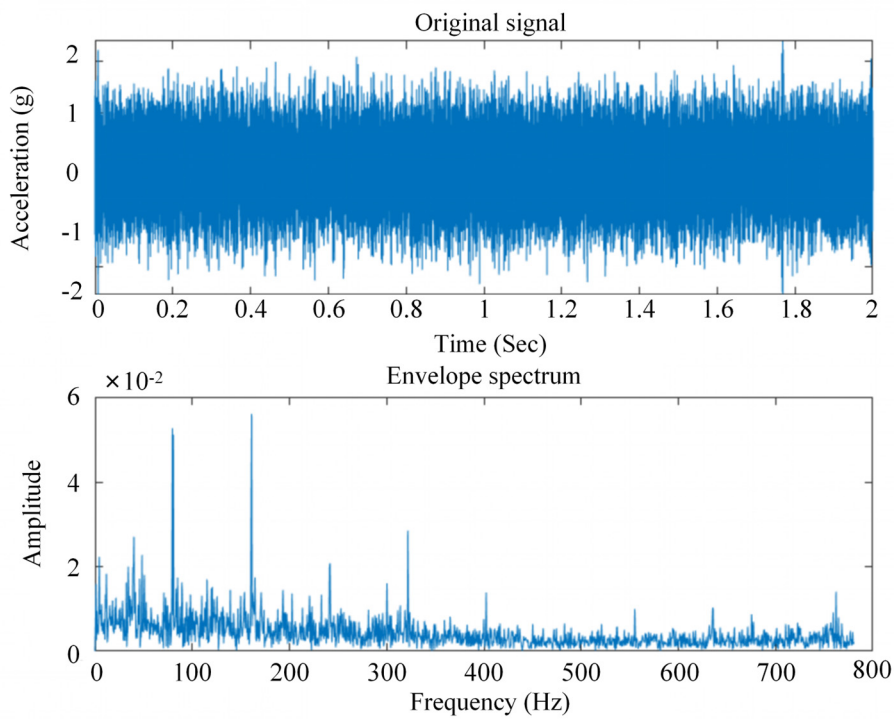


Fig. 6(f) Time domain waveform and envelope spectrum of Channel 6.

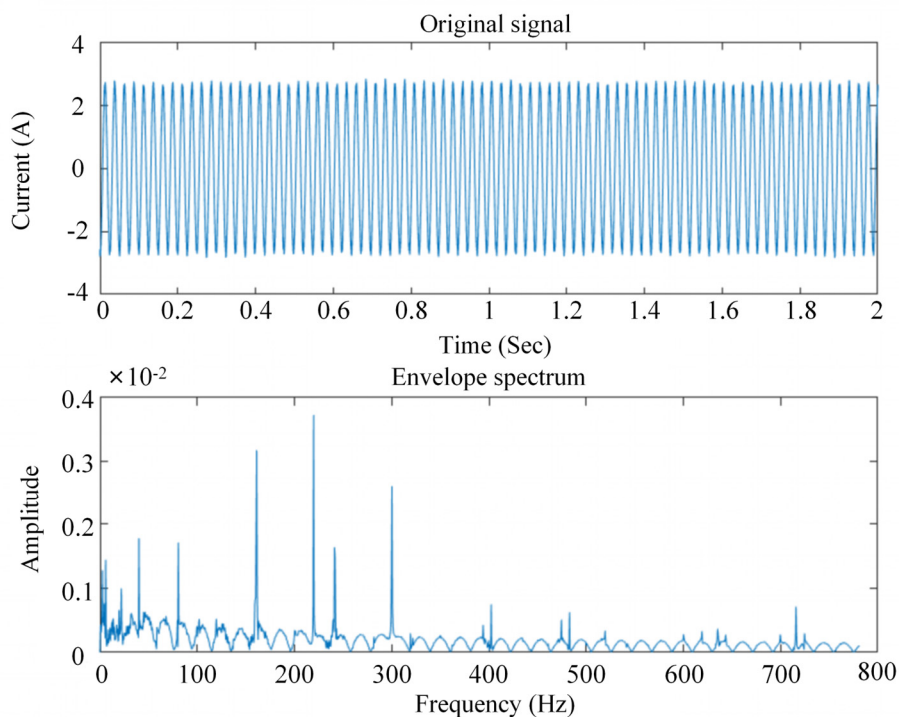


Fig. 6(g) Time domain waveform and envelope spectrum of Channel 7.

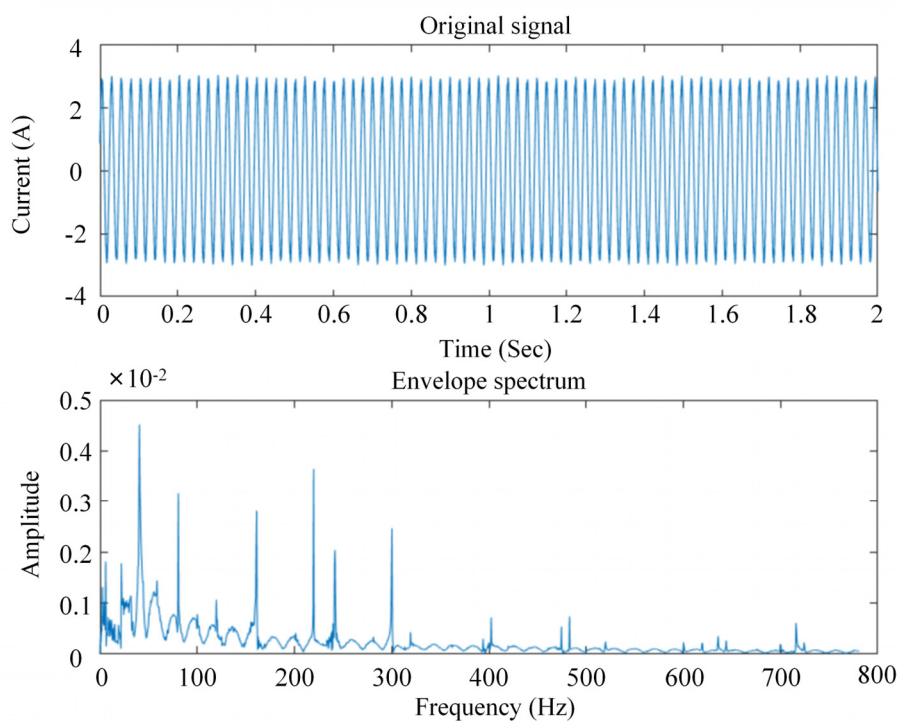


Fig. 6(h) Time domain waveform and envelope spectrum of Channel 8.

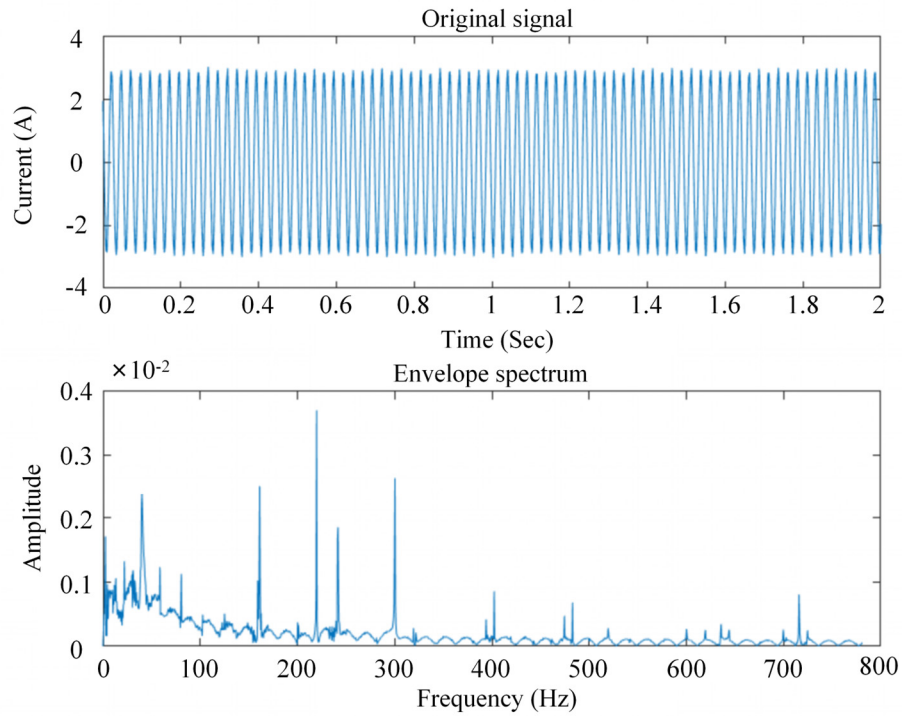


Fig. 6(i) Time domain waveform and envelope spectrum of Channel 9.

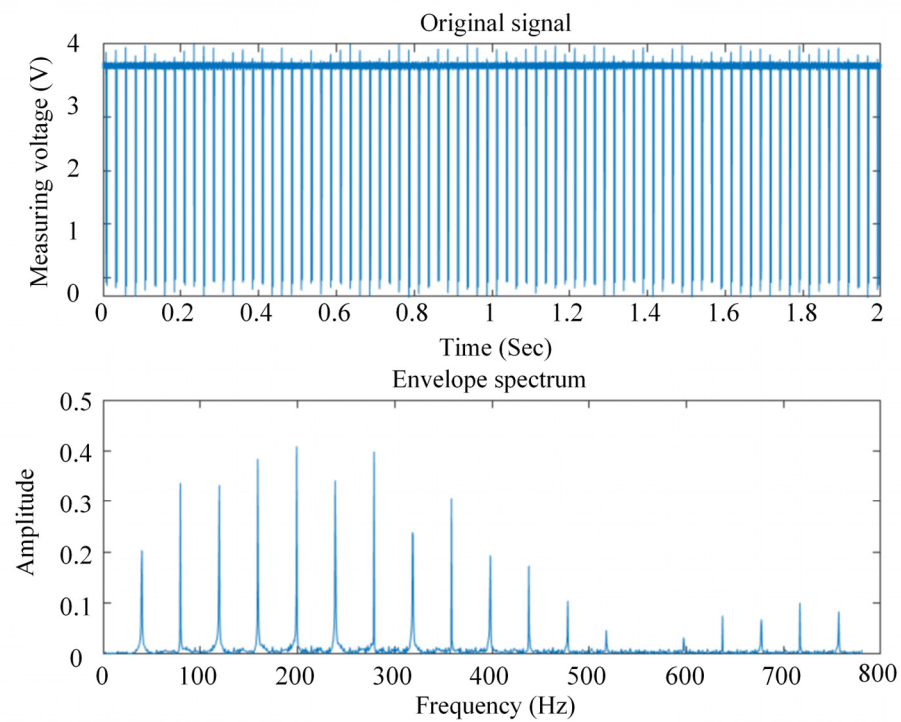


Fig. 6(j) Time domain waveform and envelope spectrum of Channel 10.

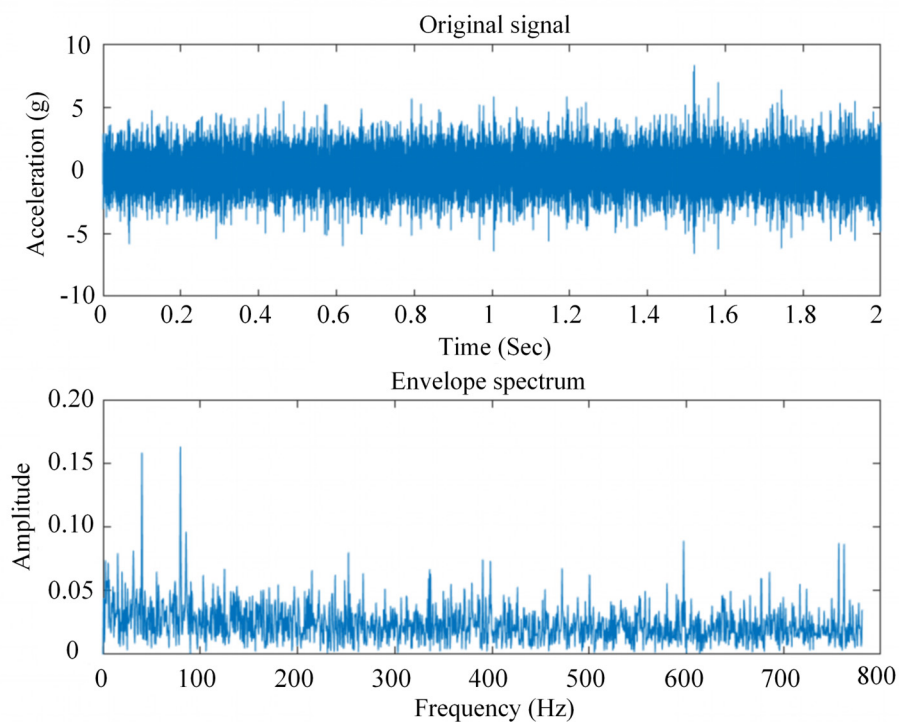


Fig. 6(k) Time domain waveform and envelope spectrum of Channel 11.

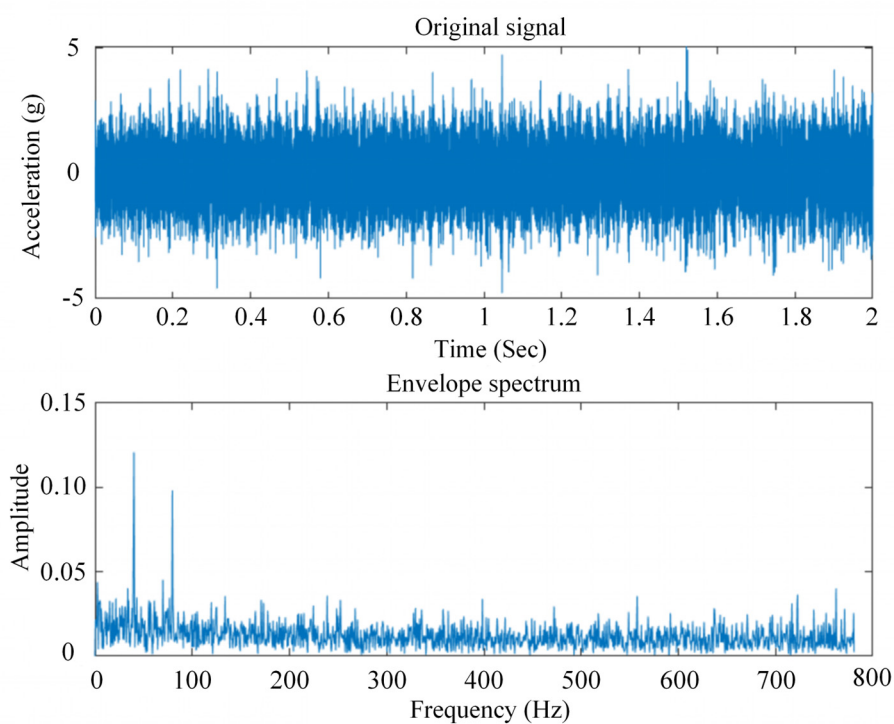


Fig. 6(l) Time domain waveform and envelope spectrum of Channel 12.

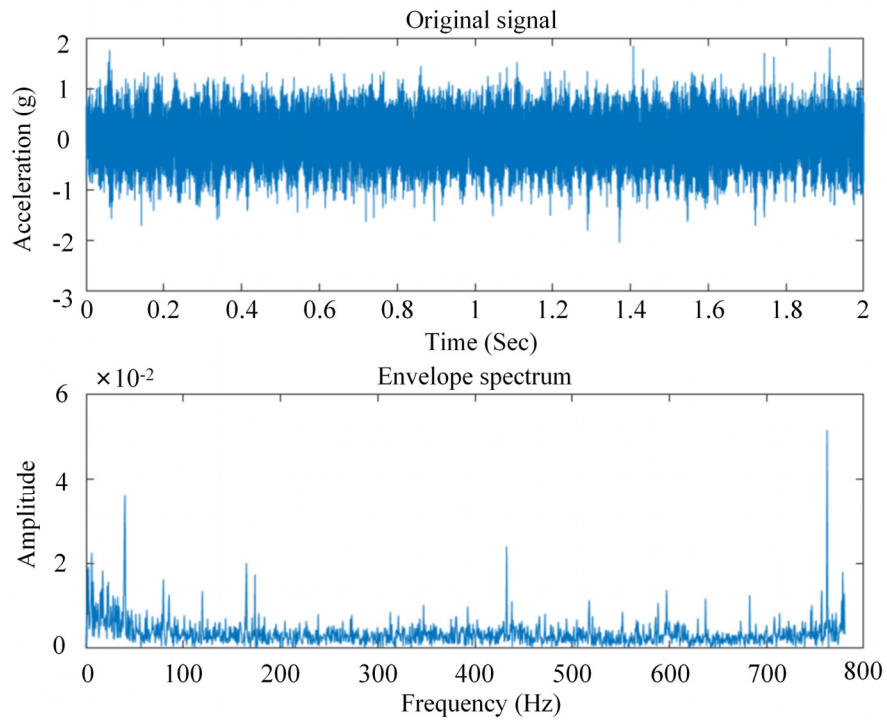


Fig. 6(m) Time domain waveform and envelope spectrum of Channel 13.

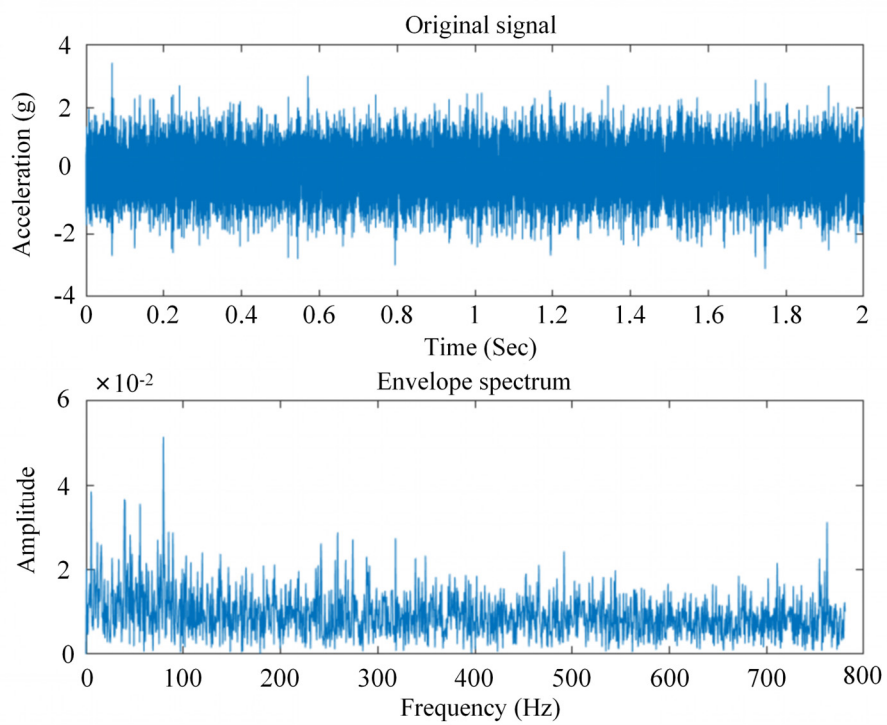


Fig. 6(n) Time domain waveform and envelope spectrum of Channel 14.

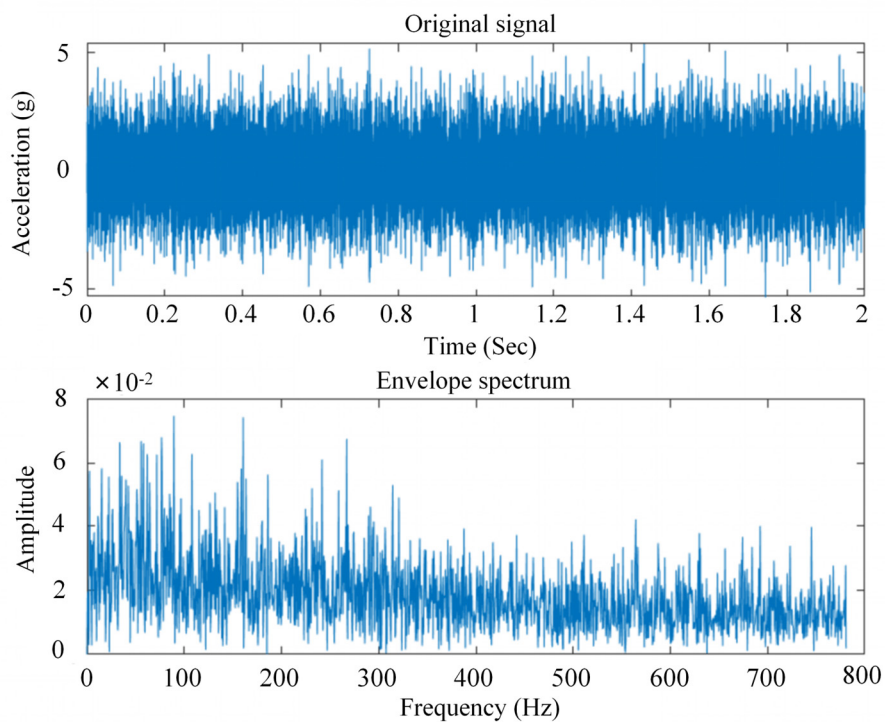


Fig. 6(o) Time domain waveform and envelope spectrum of Channel 15.

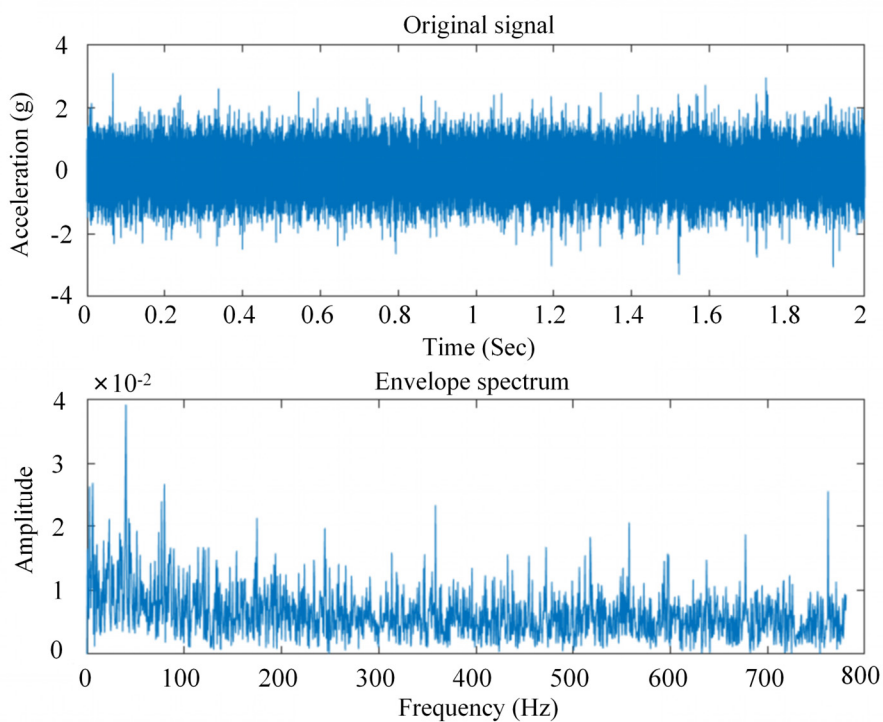


Fig. 6(p) Time domain waveform and envelope spectrum of Channel 16.

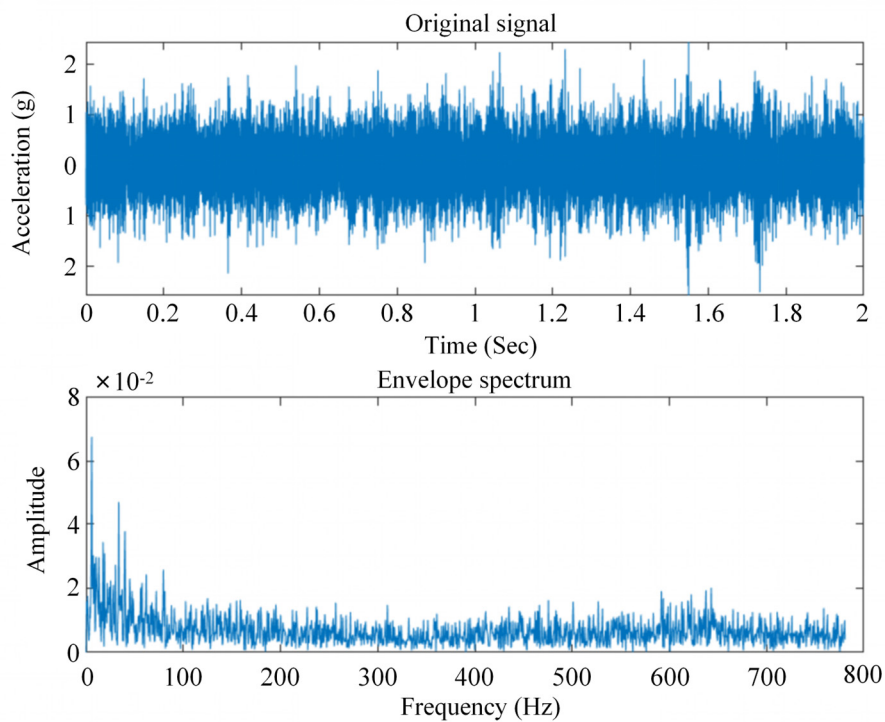


Fig. 6(q) Time domain waveform and envelope spectrum of Channel 17.

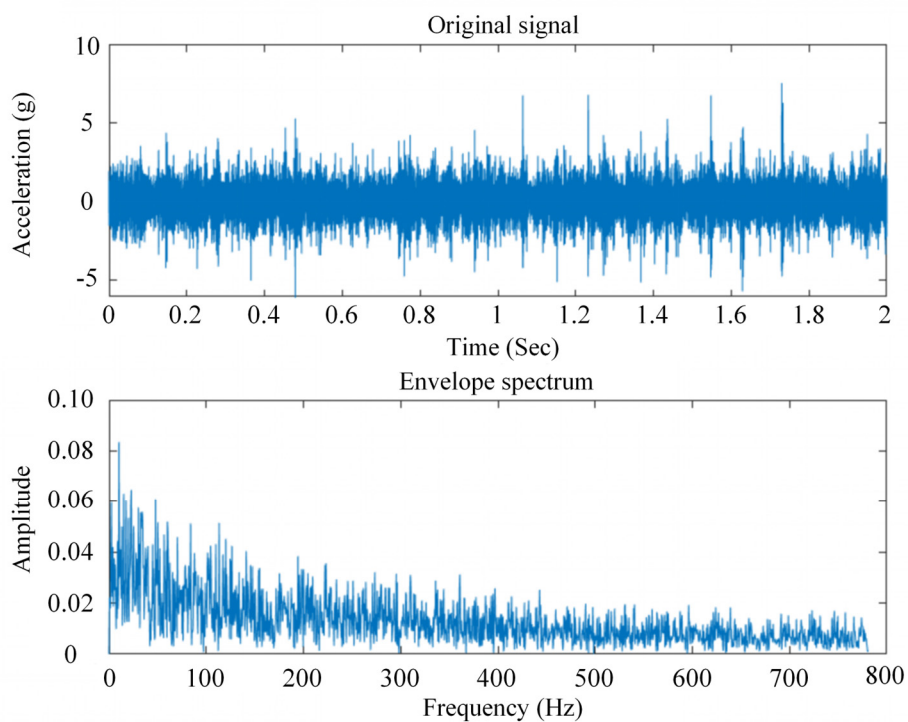


Fig. 6(r) Time domain waveform and envelope spectrum of Channel 18.

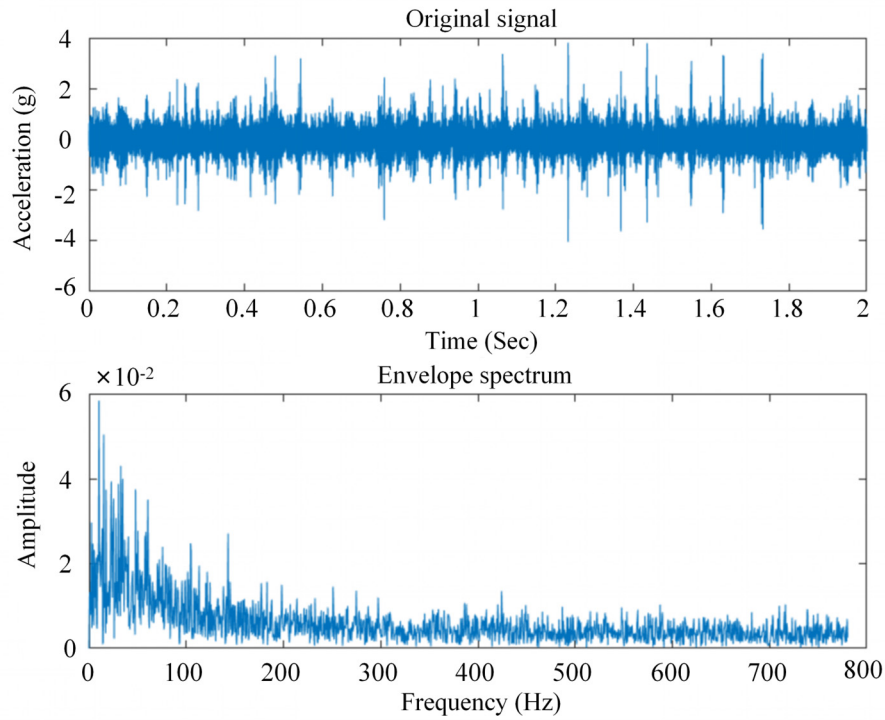


Fig. 6(s) Time domain waveform and envelope spectrum of Channel 19.

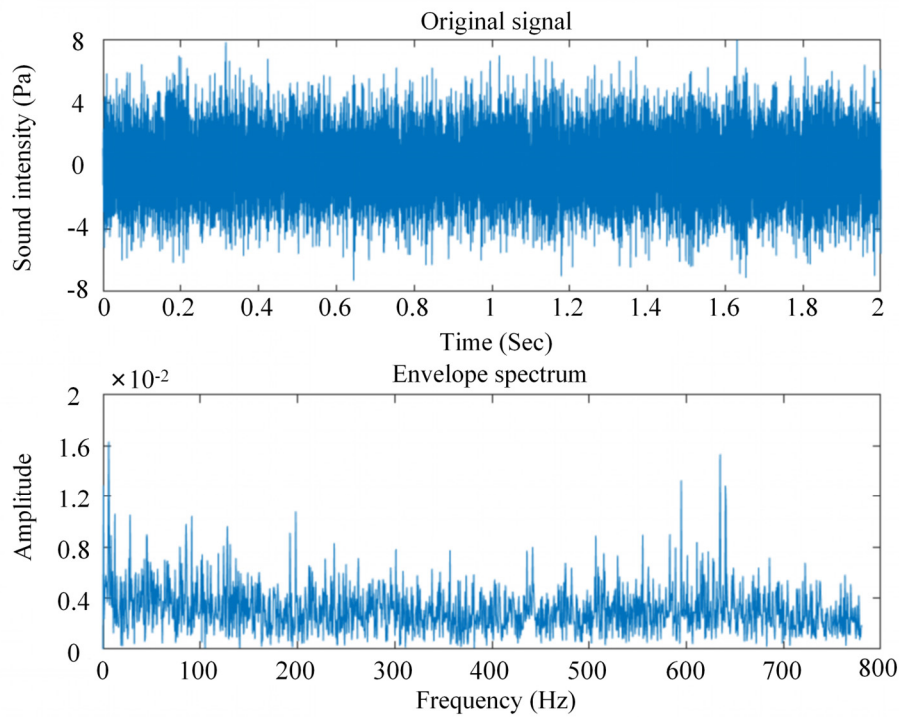


Fig. 6(t) Time domain waveform and envelope spectrum of Channel 20.

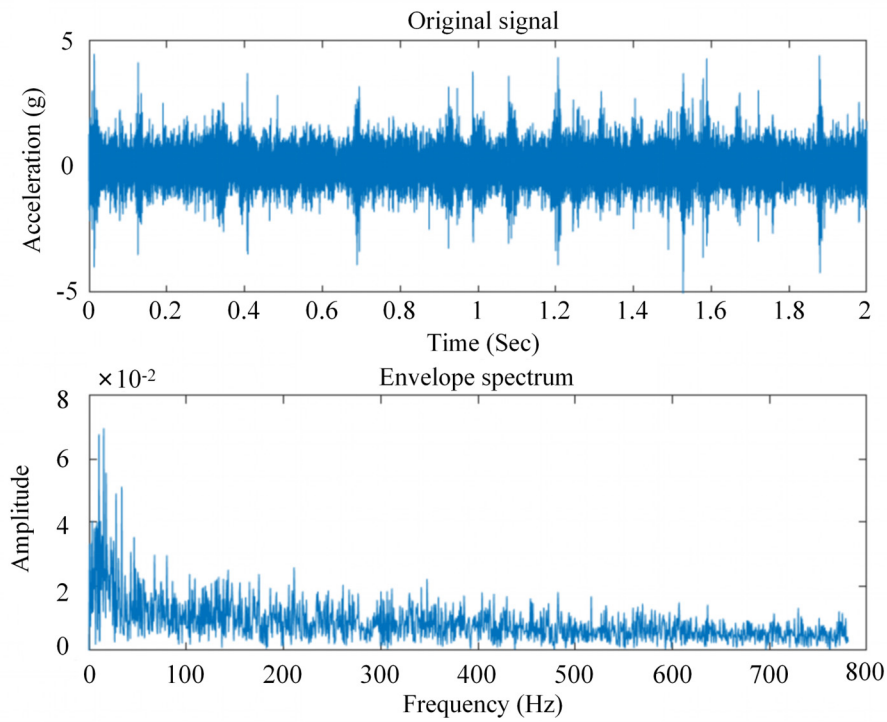


Fig. 6(u) Time domain waveform and envelope spectrum of Channel 21.

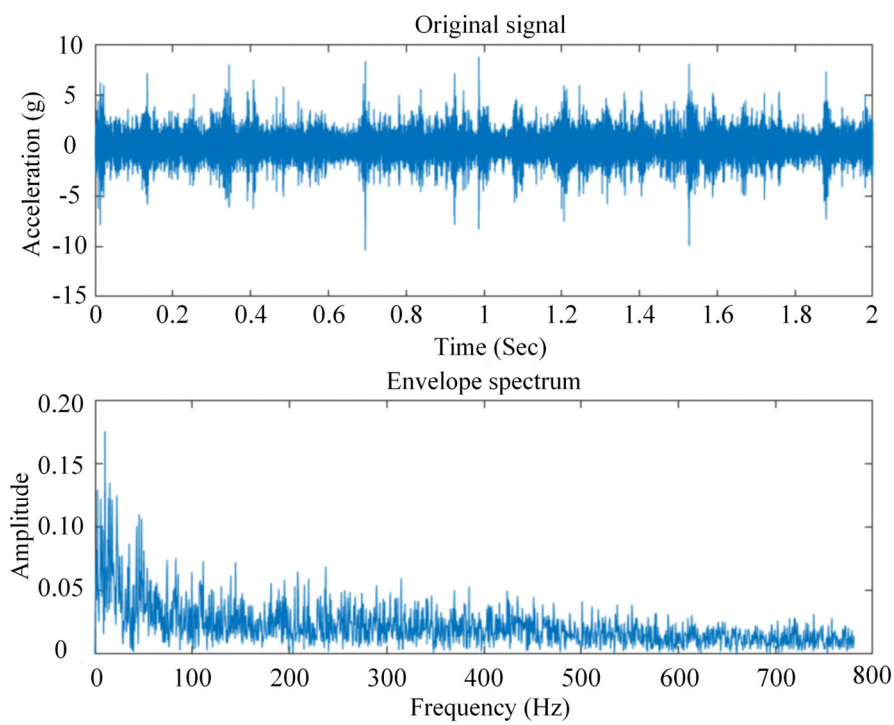


Fig. 6(v) Time domain waveform and envelope spectrum of Channel 22.

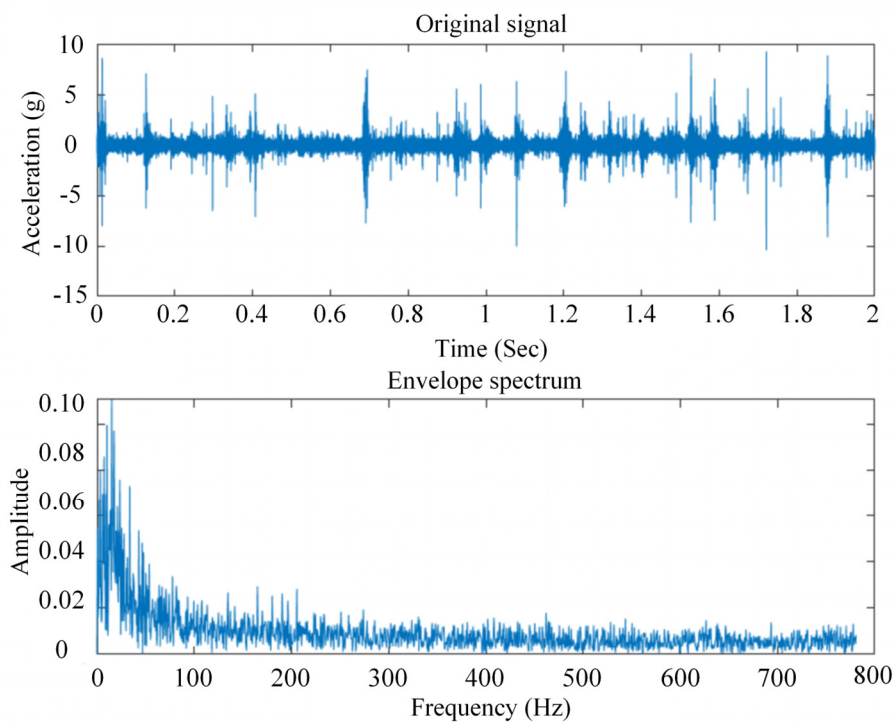


Fig. 6(w) Time domain waveform and envelope spectrum of Channel 23.

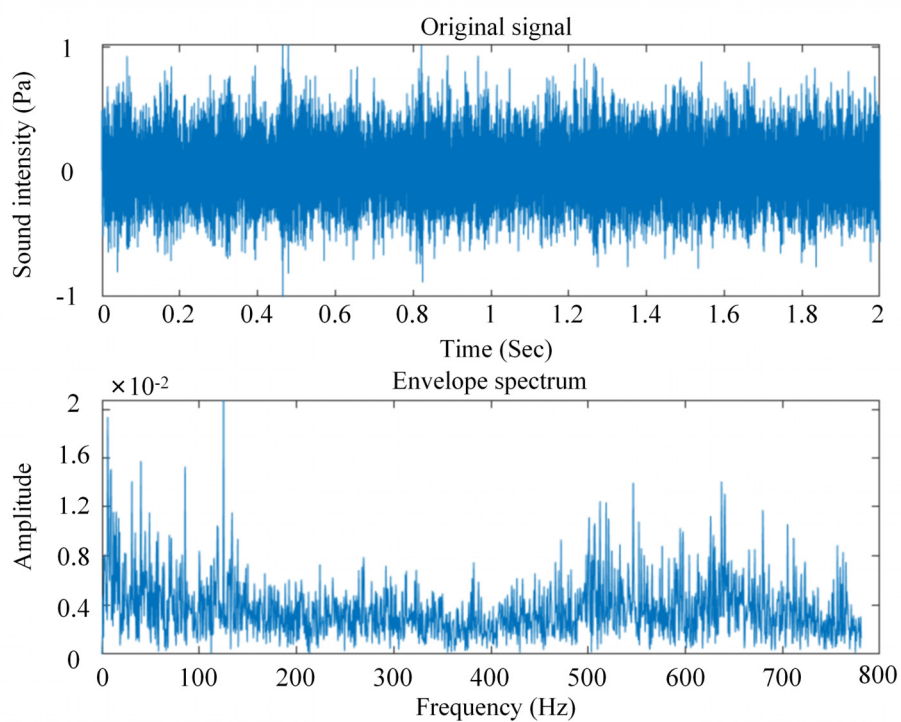


Fig. 6(x) Time domain waveform and envelope spectrum of Channel 24.

5. Download and contact

BJTU-RAO bogie datasets can be downloaded by one of the following links:

 **Baidu Netdisk:**

<https://pan.baidu.com/s/1S5F8URapHPArX2mV6FzDfA?pwd=9t74>

 **Google Drive:**

<https://drive.google.com/drive/folders/1RlZvFw-v07VvsL2Ni9cS7iFrTPDIhn2r?usp=sharing>

If you have any questions or suggestions, do not hesitate to contact:

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 **Dr. Ao Ding, e-mail: dingaobjtu@outlook.com**